

The topographic signature of mantle pressure build-up beneath subducting plates: A numerical modeling study

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Key Points:

- Positive dynamic pressure develops beneath subducting plates in subduction models within global domains.
- Slab dip is inversely correlated with the magnitude of this positive pressure.
- This pressure produces positive oceanic dynamic topography and tilts the plate upwards towards the trench.

Abstract

Subduction zones are associated with spatially heterogeneous pressure fields that, depending on location, push/pull on Earth’s surface producing dynamic topography. Despite this, subduction zones, and associated pressure fields, are typically over-simplified within global mantle flow models. Here, I use subduction models within a global domain to probe mantle pressure build-up beneath subducting plates (SPs) and the resulting dynamic topography. Positive pressure develops beneath the SP in most subduction models. This produces positive dynamic topography (≤ 450 m) and tilts the SP upwards towards the trench (≤ 0.25 m/km). As SP size increases, the magnitude of this pressure increases which, in turn, produces greater topography/tilting. At a global scale, I find potential evidence for the modeled tilting. I argue that the rigorous incorporation of subduction zones into mantle flow models, and hence the inclusion of this signal, is needed to continue to bring future dynamic topography predictions and observational estimates into closer alignment.

Plain Language Summary

Over millions of years, the solid mantle behaves as a fluid and its movement causes Earth’s outer surface to deform (producing “dynamic” topography). Computational models of this movement are now able to match the main features of observational estimates of dynamic topography, but models/observations do not match in all regions. Here, I explore whether a component of this mismatch could be due to a simplified treatment of subduction zones (i.e., plate boundaries where one tectonic plate descends beneath another) in these computational models. I create and analyze subduction models to isolate the topographic signature of subduction, focusing on topography above the subducting/oceanic plate. In the models, mantle material gets trapped beneath the plate, which pushes up on the surface and produces positive topography. This upwards-push also tilts the plate up towards the subduction zone. On average, this plate tilt appears to be present within observational estimates of Earth’s dynamic topography, but more detailed future work is needed to confirm this interpretation.

1 Introduction

The stresses associated with mantle flow produce deflections of Earth’s surface, otherwise known as “dynamic” topography. Earth’s present-day dynamic topography field can be investigated by comparing the dynamic topography generated within forward models of mantle convection (e.g., Hager et al., 1985; Ricard et al., 1993; Conrad & Husson, 2009) with the “residual” topography estimated from observed topography after removing the isostatic topography component (e.g., Panasyuk & Hager, 2000; Kaban et al., 2003; Flament et al., 2013). Recent residual topography estimates (Hoggard et al., 2016, 2017; Holdt et al., 2022), which utilize active source seismic surveys to remove the isostatic component, produce evidence for relatively short wavelength dynamic topography ($\sim 10^2$ - 10^3 km²) that must be driven by short wavelength upper mantle flow and structure (Hager & Richards, 1989; Steinberger, 2016; D. R. Davies et al., 2019; Richards et al., 2020). Such flow computations are now able to match the global characteristics of Earth’s oceanic residual topography (e.g., the relative power of long vs. short wavelength structure), but the spatial patterns of computed vs. estimated dynamic topography do not exhibit spatial agreement in certain regions. Residual topography estimates outboard of the Japan subduction zone, for example, are generally more positive than flow model predictions (e.g., Steinberger, 2016; R. D. Davies et al., 2022). Here, I use idealized models to explore the impact subduction can have on upper mantle pressure and, in turn, oceanic dynamic topography, to assess if this can help to bring future models and observations into even closer agreement.

Slabs are routinely incorporated into global flow models, but are typically not decoupled from the model surface/overriding plate (OP) or are low resolution. This restricts the dynamics of subduction (e.g., by limiting lateral motion of the trenches/slabs), which impacts mantle flow, mantle forces (pressure), and topography, and contrasts with laterally-mobile slabs in countless regional modeling studies (e.g., Christensen, 1996; Enns et al., 2005; Funiciello et al., 2004). There are notable exceptions of present-day focused global models with decoupled and high resolution subduction zones (e.g., Stadler et al., 2010; Alisic et al., 2010, 2012; Hu et al., 2022) but these studies do not examine topography. Irrespective of coupling and resolution, sinking slabs produce a dynamic draw-down of the surface (Gurnis et al., 1996; Mitrovica & Jarvis, 1985; Ricard et al., 1993; Rubey et al., 2017), as is reflected in OP residual topography (e.g., Husson, 2006; Faccenna & Becker, 2020) and sedimentary deposits (e.g., Mitrovica et al., 1989; Gurnis, 1993; Heine et al., 2008). Other second order or more local topographic features, however, do require a decoupled and well-resolved subduction zone: e.g., deep trenches, outer rises, and asymmetric across-slab dynamic topography (e.g., Zhong & Gurnis, 1992, 1994; Cramer et al., 2017; G rault et al., 2015). Regarding the latter, subduction-driven topography outboard of the trench, i.e., on the subducting plate (SP) side of the slab, has received less attention than the other topography components mentioned (cf. Husson, Guillaume, et al., 2012). This is despite having the potential to broadly affect oceanic dynamic topography due to the prevalence of wide subduction systems rimming oceanic basins. Here, I focus on this topographic component using globally-spanning subduction models which, in contrast to most previous global topography modeling studies, include decoupled and highly resolved slabs.

A static force balance, between slab buoyancy and dynamic pressure in the mantle, suggests positive pressure behind the slab (McKenzie, 1969; Stevenson & Turner, 1977) which should, in turn, produce positive SP dynamic topography:

$$(\Delta\rho gl)\cos(\theta) + \Delta P = 0 \quad (1)$$

Here, a dynamic pressure difference across the slab (ΔP) balances the slab-normal component of negative buoyancy, which is a function of the slab-mantle density contrast ($\Delta\rho$), gravitational acceleration (g), slab thickness (l), and dip (θ). The dynamic pressure difference, ΔP , connects the (indirectly) observable dip to mantle dynamics, and is defined here as the negative near-slab pressure beneath a hypothetical overriding plate (OP) minus the positive near-slab pressure beneath the SP. At depths shallower than ~ 100 km from the base of the plates, the ΔP needed to support dips $< 90^\circ$ is likely dominated by the negative pressure associated with corner flow in the mantle wedge beneath the OP (Stevenson & Turner, 1977; Tovish et al., 1978). At mid-upper mantle depths, plate and lateral slab/trench motions likely dominate (Funiciello et al., 2003; Kincaid & Griffiths, 2003), and numerical models show that ΔP becomes more evenly partitioned between positive pressure beneath the SP and negative pressure beneath the OP (e.g., A. F. Holt et al., 2017).

In summary, Equation 1 suggests that a sub-slab positive pressure signal exists for dips $< 90^\circ$ and that ΔP can be estimated observationally (from dip and oceanic plate age/density structure). This is true providing that: i) ΔP does not come solely from negative pressure above the slab, and ii), the slab geometry is relatively close to steady state (i.e., slab-normal shear forces in the slab are significantly less than ΔP). Positive pressure beneath the SP should, in turn, push on the surface and produce positive dynamic topography. Such a positive SP pressure/topography signal indeed emerges in analog subduction models, due to the confinement of mantle material behind the slab, and is suggested to be behind the relatively shallow seafloor outboard of the Scotia, Tonga, and Kermadec trenches (Husson, Guillaume, et al., 2012). However, extrapolating these results to Earth’s global range of oceanic plates sizes, and hence assessing the global relevance of this source of topography, is challenging as Husson, Guillaume, et al. (2012)

consider relatively narrow plates/trenches (1000 km) and, like A. F. Holt & Royden (2020), confine material to the upper mantle.

Observationally isolating a relatively subtle SP dynamic topography signal (below 300 m in most of our numerical single-slab models) is challenging. Such a signal could be overpowered by higher amplitude dynamic topography contributions and/or masked by uncertainties in the lithospheric isostatic correction needed to compute residual topography. I therefore turn to numerical models of subduction to isolate the nature of this sub-SP positive pressure build-up and resultant dynamic topography, in order to assess its global relevance to oceanic topography and hence guide future observational studies. Time-dependent subduction models within a global domain models are rare (Morra et al., 2009, 2010; Quevedo et al., 2013; Chamolly & Ribe, 2021; Chen et al., 2021), due to computational challenges, but enable us to incorporate a lower mantle, examine an Earth-like range of plate sizes, and avoid boundary effects. After characterizing the dependence of sub-SP pressure and dynamic topography on subduction properties, I search for the modeled topography signal within Holdt et al. (2022)’s observationally-derived residual depth estimates.

2 Methods

2.1 Modeling overview

The ASPECT finite element code (version 2.2.0) is used to construct time-evolving subduction models within 3-D global domains (Kronbichler et al., 2012; Heister et al., 2017; Bangerth et al., 2020, 2021). I investigate five global models, with equivalent mechanical properties but variable plate geometries. Four of the models contain a single subduction zone with variably sized square subducting plates (2500 km² to 11000 km²: Figure 1), chosen to represent Earth’s range of major oceanic plate sizes. For our 5th model, we consider a double subduction zone (two slabs dipping outwards from the central plate).

ASPECT was used to solve the conservation of momentum (Stokes equation) and mass (continuity equation) for an incompressible viscous fluid (Boussinesq approximation). Our models are purely compositional. That is, density and viscosity are dependent on compositional fields, which are advected through the domain using the field method (Bangerth et al., 2021). There are no external forces or velocities applied to the subduction system, and the upper and lower model boundaries are mechanically free slip.

2.2 Geometric and material properties

I use a simplified geometrical, mechanical, and rheological setup and neglect the overriding plate (cf. Morra et al., 2010; Chen et al., 2021; Chamolly & Ribe, 2021). Subduction is initiated by allowing a notch to extend to an initial depth of 150 km with a 250 km radius of curvature (Figure 1).

There are two compositional materials: the core and non-core (upper and lower) portions of the lithosphere. The 100 km thick lithosphere contains a 25 km thick core. Both lithosphere components have an equivalent density, 75 kg/m³ greater than the background mantle ($\rho_{mantle} = 3300$ kg/m³; $\eta_{mantle} = 2.5 \times 10^{20}$ Pa s), and the core is rheologically stronger ($= 2500\eta_{mantle}$) than the non-core material ($= 100\eta_{mantle}$). This is consistent with a strong core sandwiched between a brittle-yielding upper and ductile-yielding lower lithosphere (e.g., Karato & Wu, 1993). We also place stiff “core” material at the trailing edges of subducting plates to prevent subduction initiation here. The non-core lithosphere undergoes plastic yielding above a prescribed yield stress (τ_{yield}) approximated by a Coulomb friction criterion:

$$\tau_{yield} = (a\sigma_n + b)\lambda \quad (2)$$

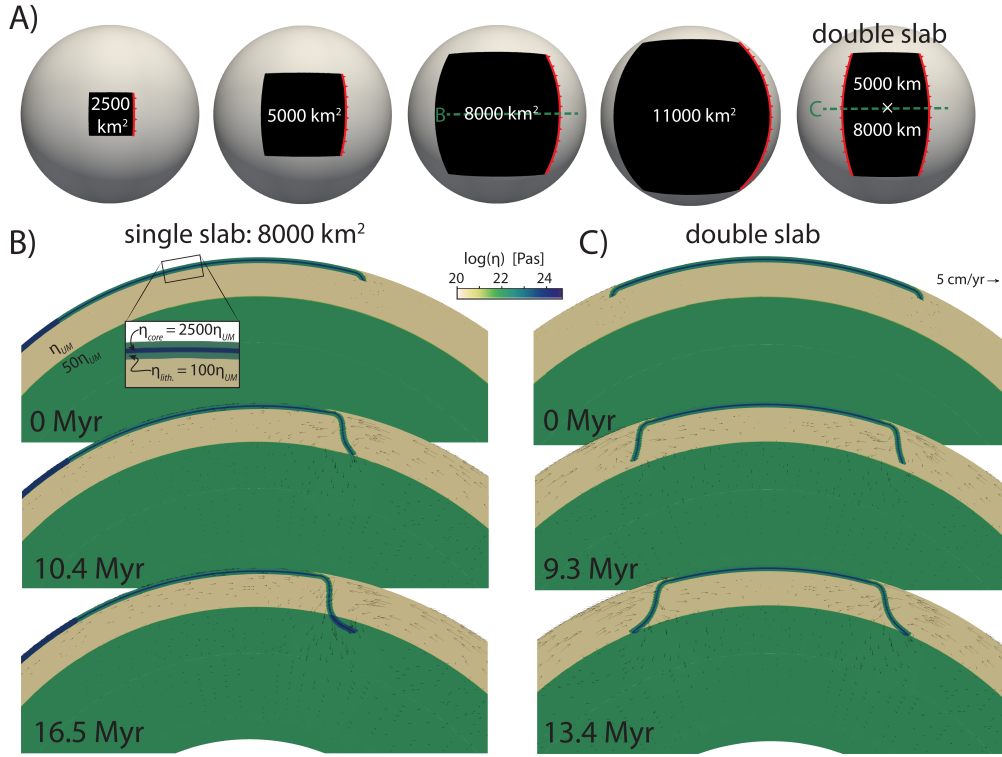


Figure 1. All initial model geometries (A) and temporal evolution of the single slab, 8000 km² SP size, model (B) and the double slab model (C). Viscosity is plotted along a vertical cross section through the center of the subduction system and overlain by velocity vectors.

a is the friction coefficient (0.6), b the cohesion (60 MPa), and λ a pore fluid factor (0.1). This λ value results in significant weakening yet does not weaken the lithosphere completely. τ_{yield} is used to calculate the plastic “viscosity”:

$$\eta_{yield} = \frac{\min(\tau_{yield}, 1 \text{ GPa})}{2\dot{\epsilon}_{II}} \quad (3)$$

where $\dot{\epsilon}_{II}$ is the strain rate second invariant. The effective viscosity (η_{eff}) is then a harmonic mean of η_{yield} and the isoviscous component. This produces averaged non-yielding lithosphere viscosities of $\sim 200 \eta_{mantle}$ and yielding viscosities of $\sim 100 \eta_{mantle}$. Such moderate slab strengths are consistent with low elastic thicknesses at the trench (Billen & Gurnis, 2005), tomographically imaged slab morphologies indicating low viscosity deformation (e.g., Čížková et al., 2002), and plate driving force/slab dynamics considerations (e.g., Wu et al., 2008; Funiciello et al., 2008). The background mantle viscosity increases by factor 50 at 660 km depth.

2.3 Numerical parameters

Adaptive mesh refinement (AMR) occurs for finite elements with large gradients in viscosity and/or non-zero compositions. This enables us to highly resolve the lithosphere while capturing mantle-scale flow. The AMR parameters produce a maximum refinement corresponding to ≈ 5.5 km radially-thick finite elements (in the lithosphere) and a minimum refinement corresponding to ≈ 90 km elements (lower mantle). I have conducted tests to ensure our solver tolerances are sufficiently strict, and that upper mantle pressure fields are minimally affected by lower mantle resolution which, when low, can produce pressure artifacts that are restricted to the lower mantle (Figure S1).

2.4 Model analysis

To isolate links between pressure, topography, and subduction properties, the following quantities are extracted from the models:

- *Slab dip*: Dip is computed at 250 km depth using the location of the slab upper surface at 50 km above and below this depth.

- *Dynamic pressure*: Dynamic pressure values in the upper mantle (i.e., the non-lithostatic component of full pressure), also at 250 ± 50 km depth, are extracted from beneath both the subducting plate (SP) and a hypothetical overriding plate (OP). At each depth, P_{SP} is extracted at 20 km from the lower surface of the slab (i.e., positive sub-slab pressure), P_{OP} is extracted at 20 km from the upper surface (negative mantle wedge pressure), and $\Delta P = P_{OP} - P_{SP}$ is computed. Our final ΔP is depth-averaged: $\Delta P = 0.25(\Delta P_{200 \text{ km}} + 2\Delta P_{250 \text{ km}} + \Delta P_{300 \text{ km}})$.

- *Dynamic topography*: The radial normal stress at the upper model boundary is used to estimate topography ($DT = \sigma_{rr}/\Delta\rho g$) (e.g., Zhong & Gurnis, 1994), where $\Delta\rho$ is 2300 kg/m^3 (water-loaded topography). The constant ‘isostatic’ topography component, due to the weight of the uniformly thick SP, is subtracted to leave ‘dynamic’ topography. I extract topography at 500 km from the trench, to be close to the slab yet avoid short-wavelength flexural topography, and also at 800 km to compute the topography gradient.

3 Results and Discussion

3.1 General model evolution

The four single slab subduction models evolve similarly. Firstly, the slabs sink through the upper mantle with 6-8 cm/yr vertical sinking rates (convergence is $\approx 50\%$ rollback, 50 % plate motion) and 60-70° dips. After ~ 10 Myrs, the slabs impinge on the upper-

to-lower mantle boundary, are bent, and sink slowly through the lower mantle (Figure 1). This later phase is associated with greater dips ($70\text{--}80^\circ$), particularly for the largest plate (11000 km^2) case. In the double subduction model, convergence is accommodated almost purely by rollback and dips are reduced ($\approx 45\text{--}60^\circ$).

I focus on model times between ≈ 7 and ≈ 20 Myr. This ensures the slab is deep enough to extract a dip yet avoids the latter-most stages during which subduction sometimes initiates at the plate edge. This could be avoided by placing a flat plate adjacent to the SP (e.g., Yamato et al., 2009); Here, I simply focus on the earlier phases.

3.2 Links between subduction properties, dynamic pressure, and topography

Upper mantle pressure: In all models, and during most timesteps analyzed, positive pressure develops behind (i.e., beneath the SP) and negative pressure develops ahead of retreating slabs (i.e., beneath the OP) at mid-upper mantle depths (Figure 2). On the SP-side, positive pressure build-up is mainly driven by SP motion (i.e., in order to drive flow in the opposite direction to the SP and hence conserve mass) and slab rollback (i.e., due to compression of the sub-slab mantle material). On the OP-side, negative pressure is also induced by rollback, with an additional contribution from “corner flow” in the mantle wedge. This wedge-directed decrease in pressure produces toroidal flow around the slab edge (Dvorkin et al., 1993; Funicello et al., 2003; Kincaid & Griffiths, 2003) and a normal force on the slab that produces dips $< 90^\circ$ (Stevenson & Turner, 1977; Tovish et al., 1978).

In single slab models, the magnitude of sub-slab positive pressure increases with plate size (Figures 2; 3a; S2). This agrees with analytical expressions linking greater sub-slab pressure with larger plates/wider slabs (Royden & Holt, 2020). Of all the models, sub-slab pressure becomes near-zero only during the final stages of the small SP (2500 km^2) model’s evolution. In the double slab model, sub-slab pressure values are $\approx 65\%$ greater than the single slab model with equivalent trench width. This is due to the confinement of mantle material between the two slabs (cf. A. F. Holt et al., 2017; Király et al., 2018) and a $\sim 50\%$ increase in slab rollback relative to the single slab models (Figure S2).

Slab dip: I now consider the imprint of sub-SP pressure build-up on dip and topography. Equation 1 shows how across-slab pressure difference (ΔP ; Figure 3d) links to dip (θ) via a static force balance. For both single and double slab systems, and variable plate sizes, the pressures and dips conform to this relationship to first order (Figure 3a), albeit with a 5–10 MPa offset (likely due to non-steady state effects, i.e. internal slab shearing). That is, as plate size increases or an extra slab is inserted, the pressure magnitude increases, which supports shallower dips (cf. A. F. Holt et al., 2017). During most timesteps, positive sub-slab pressure (P_{SP}) contributes to at least half of the across-slab pressure difference, $\Delta P (= P_{OP} - P_{SP})$, that supports slab dip. Hence, models have large-variability in P_{SP} , which decays to near-zero towards the SP edge (Figure 2).

Dynamic topography: Positive pressure build up produces positive SP dynamic topography in the subduction models (Fig. 3b, S3). This was explored within Husson, Guillaume, et al. (2012)’s analog models, and can also be seen in Crameri et al. (2017)’s numerical models. In our models, the magnitude of this topography reaches $\sim 450\text{ m}$ in the double slab case. In most of the single slab models, topography peaks at 100–300 m. Dynamic topography is maximum at the center of the subduction system where pressure build-up is greatest (cf. Figs 2, S3). There are only a few timesteps during which SP topography is negative (Fig. 3b), which correspond to the latter stages of the smaller plate model runs. Following the negative correlation between sub-SP pressure and dip (Figure 3a), there is a similar negative correlation between dynamic topography and dip (Figure 3b). This is because positive mantle pressure dictates both properties (i.e., by

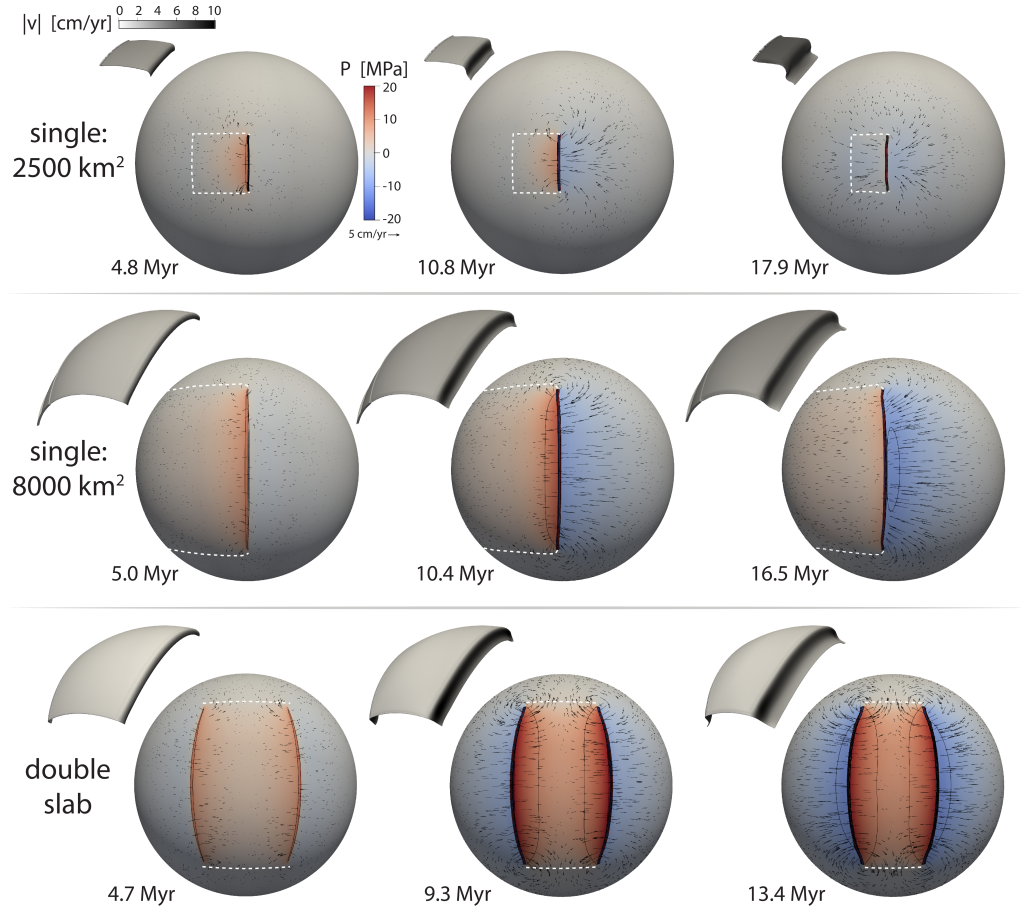


Figure 2. Dynamic pressure evolution within two single slab subduction models (2500 km² and 8000 km² SP sizes) and the double slab model. Snapshots show the dynamic pressure field and mantle velocity vectors at 330 km depth, the surface location of the SP (dashed white line), and insets of the plate and slab shape (shaded by velocity magnitude). The dynamic pressure is normalized to produce zero average pressure at this depth.

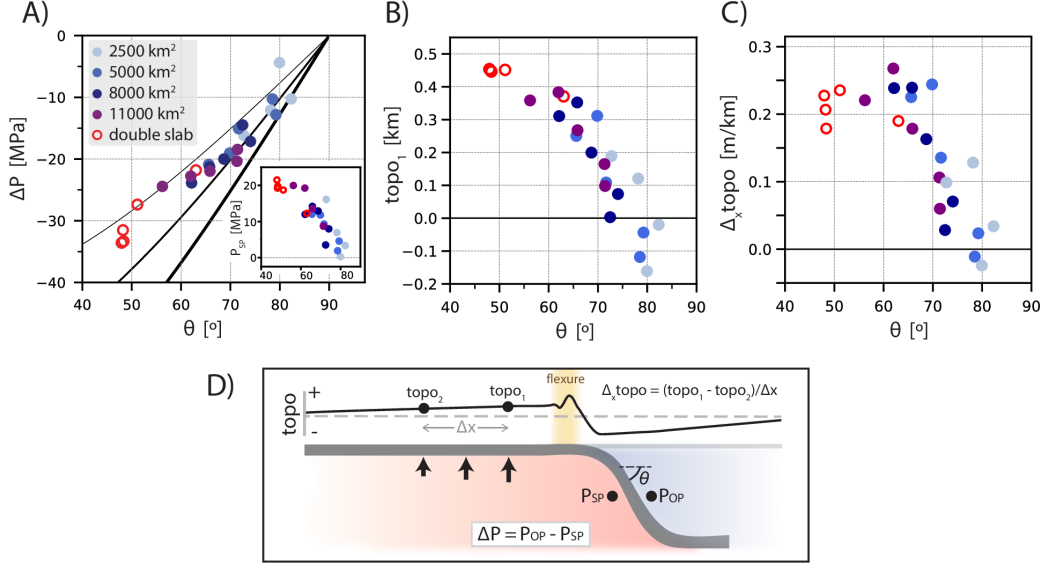


Figure 3. Dynamic pressure and subducting plate topography as a function of dip: A) Across-slab dynamic pressure difference (ΔP) and sub-slab dynamic pressure (P_{SP}) vs. dip (θ); B) subducting plate topography ($topo_1$) vs. θ ; C) subducting plate topography gradient ($\Delta_x topo$) vs. θ . Panel A) includes the force balance of Equation 1 for ΔP balancing the full slab buoyancy (thick line), 80% (medium line), and 60% of the buoyancy (thin line). Panel D) schematically illustrates the locations of extracted topography and pressure.

pushing slabs/Earth's surface). As for pressure, SP topography decreases from the sub-slab to the trailing edge causing an upwards tilt towards the trench (Figure 3c,d). This trench-ward tilt reaches ≈ 27 m per 100 km (or 0.015°) and extends to a few-hundred km from the trench (where bending-related topography becomes dominant). While draw-down/tilting of upper plates is well established (e.g., Mitrovica et al., 1989, 1996), and often of greater amplitude (e.g., $\sim 0.08^\circ$ of Eocene tilt for Australia: DiCaprio et al. (2009); Heine et al. (2008)), I am unaware of previous work focusing on subduction-driven SP tilting.

3.3 Modeling limitations

Despite enabling us to extract first-order relationships, the models are highly simplified. Most notably, the overriding plate (OP) is neglected. This is due to resolution limitations but in line with other time-dependent subduction models within a global domain (Morra et al., 2009, 2010; Chamolly & Ribe, 2021; Chen et al., 2021). Including an OP would reduce the rate of trench retreat relative to SP motion (Yamato et al., 2009; Butterworth et al., 2012) and shift the mantle wedge corner (and hence negative pressure associated with corner flow) to greater depth. This would result a ΔP signal that, while dictated by slab buoyancy (Equation 1), may be more dominated by negative P_{OP} (at the expense of positive P_{SP}).

Other simplifications include the mantle rheology and density structure. Including the power-law component of mantle viscosity, which we neglect, would reduce the suction force associated with corner flow (Tovish et al., 1978; Billen & Hirth, 2005) and shear stresses acting on the subducting plate (e.g., Jadamec & Billen, 2010). This pro-

duces steeper dips (Billen & Hirth, 2005) and reduced trench retreat (e.g., A. Holt & Becker, 2017), and so could potentially offset enhanced rollback caused by neglecting the OP.

Flow driven by non-slab density anomalies, other subduction zones, deep slab material accumulated in the lower mantle, and the motions of other plates is also neglected. The pressure patterns associated with such far-field flow will affect dynamic topography around subduction zones, particularly dragging down the surface above accumulated slab material (e.g., Pysklywec & Mitrovica, 1998; Shephard et al., 2012; Rubey et al., 2017), and subduction properties like dip (e.g., Ficini et al., 2017; Chertova et al., 2018; Husson, Conrad, & Faccenna, 2012; A. F. Holt & Royden, 2020).

3.4 Subducting plate topography on Earth

I now investigate whether such topographic signals exist at Earth subduction zones using Holdt et al. (2022)’s observationally-derived residual topography (RT). An update of Hoggard et al. (2016, 2017), this dataset uses active-source seismological datasets to correct for sedimentary and crustal loading, a plate cooling model to estimate and remove ocean lithosphere loading, and comprehensively covers the ocean floor (Fig. S4a). I compare this subduction zone topography (Fig. 4a) with dynamic topography (DT) predictions from two recent convection models (Figs. 4b, S4c,d) (Steinberger et al., 2019; R. D. Davies et al., 2022). Earth subduction zones are split into ~ 220 km-long segments, with segments associated with ridges, plateaus, or continental material omitted (cf., Lallemand et al., 2005). For Holdt et al. (2022)’s dataset, I also consider a case where segments without ≥ 5 nearby residual depth spot measurements are thrown out. This is to check the results are independent of the algorithm used to generate a continuous RT field. For each dataset, I assess whether observationally-derived (RT) or modeled (DT) topography, and/or its gradient, is positive above these SP segments.

At a global scale, I do not find evidence for preferentially positive RT above SPs (Fig. S5a). The mean RT is weakly negative irrespective of whether all or the filtered subset of subduction segments are considered (-0.16 to -0.07 km). This is unchanged when long wavelength structure (spherical harmonic degrees < 10) is removed from the global grid (Fig. S6a). The mean topography is more negative in the DT models: -0.56 km to -0.31 km (Steinberger, 2016; R. D. Davies et al., 2022). The positive SP topography we model would, in principle, reduce this mismatch but this is speculative as the RT/DT distributions overlap significantly. This lack of evidence for an SP DT signal may be due to overprinting of this few-hundred meters signal by larger amplitude contributions. Within global geodynamic models, the gradual sinking of relic slabs from previous episodes of subduction - particularly those located in the lower mantle - have been shown to draw-down subduction zones by heights on the order of a km (e.g., Hager, 1984; Ricard et al., 1993; Conrad & Husson, 2009). Our slabs are mainly contained within to the upper mantle (Figure 2) and so we do not include such effects. Also, calculating RT above old ocean floor relies on semi-empirical corrections for ocean plate loading that may also mask the modeled topographies (e.g., Parsons & Sclater, 1977; Stein & Stein, 1992; Holdt et al., 2022). The modeled topography may also be associated with a gravity signal: For the mass excess due to 200 m of positive topography, approximated as a half-ellipsoid, simple calculations predict a maximum gravity signal of 20-35 mGal at the center of the anomaly (Fig. S7a). Such a signal is broadly compatible with the positive free-air gravity observed at some subduction zones (e.g. Central and South America, Aleutians, Kuril), but this is not universal and requires more detailed analysis/modeling to untangle this signal from that due to excess slab mass.

I now investigate the modeled positive topography gradient: An upwards tilt of the SP towards the trench. Figure 4 shows the distribution of $\Delta_x \text{topo}$ for Earth subduction zones, extracted from each topography map. Observational estimates, from the RT map, are more positive than the topographic gradients predicted by mantle flow models. Sub-

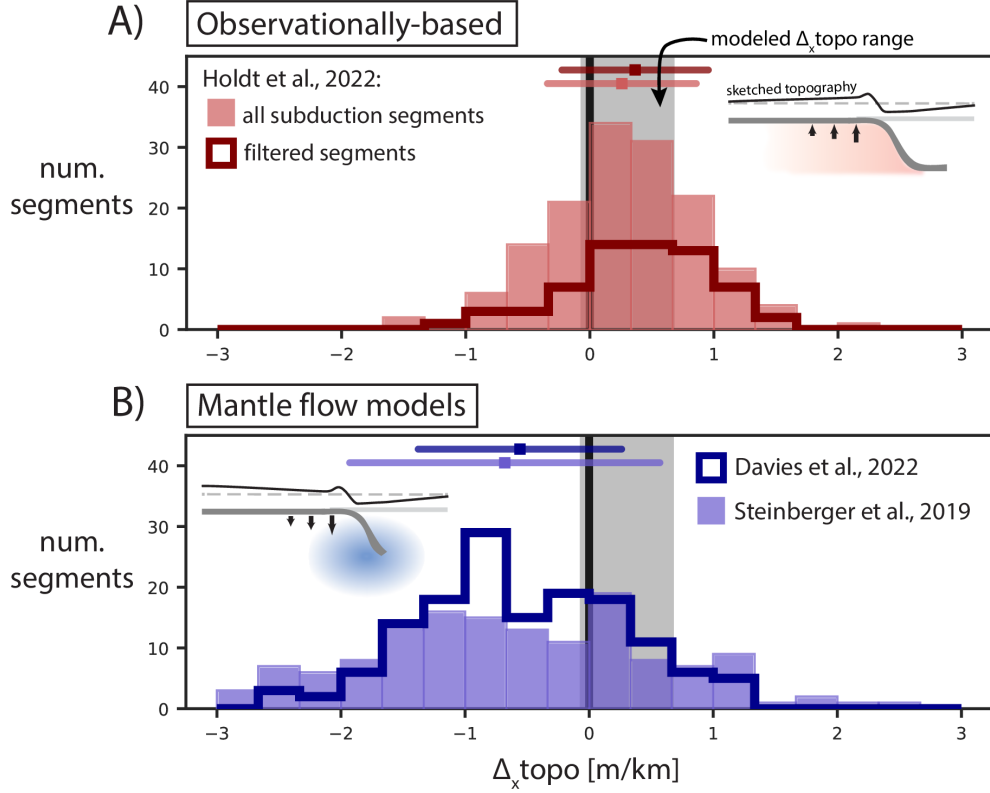


Figure 4. The distribution of subducting plate topography gradient within published: A) observationally-based residual topography (Holdt et al., 2022); B) dynamic topography models from mantle flow computations (R. D. Davies et al., 2022; Steinberger et al., 2019). Topography gradient ($\Delta_{x,topo}$) is extracted 400 km outboard of the trench, and positive values represent upwards tilt towards the trench. Horizontal bars show the arithmetic means plus/minus the standard deviation. The gray bar represents the range of $\Delta_{x,topo}$ values in our subduction models (Fig. 3c), scaled up according to the difference between the modeled ΔP range (~ 30 MPa) and the “observed” range calculated from Eq. 1 (~ 75 MPa). The Steinberger (2016) topographies are expanded up to spherical harmonic degree 31 (and is their case without lateral viscosity variations) and Holdt et al. (2022) up to degree 41. R. D. Davies et al. (2022) is their case with both upper mantle structure and temperature/pressure-dependent viscosity. In the “filtered” version of Holdt et al. (2022), subduction segments are counted only if they are close (≤ 150 km) to 5 or more residual depth measurements.

duction zones within Holdt et al. (2022)’s dataset exhibit a mean upwards tilt of 0.25 m/km, and this increases to 0.37 m/km when only subduction segments closer to ≥ 5 observations are included (Fig. 4a). In contrast, the DT models exhibit negative/downwards average tilts of -0.68 to -0.56 m/km (Fig. 4b). The average tilt in the RT dataset reduces when topography is extracted further from the trench (Fig. S8), but $\Delta_x \text{topo}$ is always shifted to more positive values relative to the DT simulations. This is also the case when comparing Steinberger et al. (2019)’s DT to the RT maps generated within that paper using a global, Crust1.0-based crustal (Laske et al., 2013). Steinberger’s mean tilt is ~ 1 m/km more positive within the RT models, irrespective of whether ocean plate loading is removed from the RT/DT maps or not (Fig. S9). I propose this mismatch is evidence for an upwards tilt of SPs that, despite large RT scatter (due to the dominance of other topographic sources), is not reflected in flow models without decoupled slabs. The negative tilt in mantle flow models is due to the dynamic drawdown driven by dense slab anomalies, in contrast to the dominance of sub-SP pressure in our subduction models. This drawdown is accentuated in older flow computations (Conrad & Husson, 2009; Spasojevic & Gurnis, 2012) that are more dominated by deeper and larger-scale anomalies from older tomographic models (Fig. S10).

4 Conclusions

I have used time-dependent subduction models, within a global domain, to elucidate links between subduction properties, mantle pressure, and SP dynamic topography. The magnitude of upper mantle dynamic pressure, which is positive behind and negative ahead of slabs, increases with plate size, slab length, and trench velocities. This agrees with previous analytical and numerical subduction models (e.g., Royden & Husson, 2006; Stegman et al., 2006) but extends this framework globally. As across-slab pressure difference (ΔP) increases, e.g. with increasing plate size, dip decreases due to an increased mantle-wedge directed pressure force. This is approximated by a force balance that demonstrates how ΔP can be estimated from dip (cf. A. F. Holt & Royden, 2020).

The pressure distribution produced by the confinement of mantle material behind a slab - i.e., positive behind the slab and decaying with slab-normal distance - produces positive SP dynamic topography and causes the plate to tilt upwards towards the trench (at distances beyond a few-hundred km from the trench; here, flexure dominates). For a Pacific Plate sized plate ($\sim 11000 \text{ km}^2$), positive dynamic topography of 100-400 m is produced close to the slab. This is reduced to -100-300 m for a Nazca sized plate ($\sim 5000 \text{ km}^2$). In both cases, a trench-upwards tilt of ≤ 0.25 m/km is recorded.

SPs within Holdt et al. (2022)’s residual topography (RT) dataset exhibit tilts that are, in general, more positive than dynamic topography (DT) extracted from recent convection models. I interpret this as evidence that DT simulations, which do not contain decoupled slabs, are missing a component of Earth’s dynamic topography related to sub-slab and sub-SP pressure build-up. Unlike tilt, I do not find evidence for positive absolute topography above SPs at a global scale. This is unsurprising as sinking slab remnants, and/or RT uncertainties, could offset this relatively subtle (few-hundred meters) signal. Moving forward, geographically-accurate global subduction models are needed to confirm and parse out the contribution of this source of oceanic DT.

Open research

All ASPECT model inputs are downloadable from the following GitHub repository: <https://github.com/AdamFHolt/global-subduction-ASPECT>. They will be transferred to a permanent Zenodo repository upon (potential) paper acceptance.

Acknowledgments

This work has benefited greatly from discussions with Leigh Royden. Samuel Goldberg and Thorsten Becker are also thanked for helpful discussions and assistance processing the topography datasets. The computations used the Extreme Science and Engineering Discovery Environment (XSEDE), which is supported by National Science Foundation (NSF) grant number ACI-15485x62. AFH was partially supported by NSF grant EAR-2147997. I also thank the Computational Infrastructure for Geodynamics (geodynamics.org), which is funded by the NSF under awards EAR-0949446 and EAR-1550901, for supporting the development of ASPECT.

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